Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State what is meant by a member in tension.

- The member is being pulled / stretched by the forces at its ends

3 Quick check

- a) CWM = ACWM
- b) 100 N m
- c) Strut
- d) 980 N

4 Key terms

- 1 – B
- 2 – F
- 3 – C
- 4 – E
- 5 – A
- 6 – D

5 State the two conditions of equilibrium.

- The sum of the forces = 0 (upward = downward, left = right)
- The sum of the moments = 0 (clockwise = anticlockwise)

6 Calculate the reactions at each support.

- F = mg = 95 × 9.8 = 931 N
- Moments about the left support: 931 × 9 = R_right × 25
- R_right = 335 N
- R_left = 931 − 335 = 596 N